# PAPER_665: UQFF Hawking Suppression Equations

**Subtitle:** Derives the chain of multiplicative suppression factors S1, S2, S3 that reduce Hawking luminosity in UQFF, with sensitivity analysis. **Module:** UQFFSuppressionEquationsHawking
**Session:** Session 172
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## Abstract

Three UQFF suppression factors reduce Hawking radiation luminosity multiplicatively. We derive each factor analytically, combine them into a total suppression S_total, and perform sensitivity sweeps over the aether density ratio.

## 1. Suppression Factors

### S1 — Negentropic Modulation (affects T_H only)

$$S_1 = 1 + f_{TRZ} = 1.1$$

### S2 — [UA]/[SCm] Density Ratio

$$S_2 = 1 - \frac{\rho_{SCm}}{\rho_{UA}} = 0.9$$

### S3 — Magnetic String Exponential

$$S_3 = \exp\left(-\frac{U_m}{k_B T_H}\right)$$

## 2. Modified Quantities

$$T_{UQFF} = T_H \cdot S_1 \cdot S_2$$

$$L_{UQFF} = L_H \cdot S_2 \cdot S_3$$

$$\frac{dT_{UQFF}}{dM} = \frac{dT_H}{dM} \cdot S_1 \cdot S_2$$

## 3. Total Suppression

$$S_{total} = S_1 \cdot S_2 \cdot S_3$$

For U_m/k_BT_H = 1: S_total ≈ 1.1 × 0.9 × 0.368 ≈ **0.364**

Hawking luminosity reduced to ~36% of GR prediction.

## 4. Sensitivity Sweep

As ρ_UA/ρ_SCm varies from 2 to 20: S2 ranges 0.5→0.95, strongly affecting L_UQFF.

## 5. C++ Module

`UQFFSuppressionEquationsHawking.h / .cpp` — Session 172 CP4 #249 — `UQFFSuppressionEquationsHawkingCalculator`

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